Mutual Fund Analytics

Capstone Project – Final Report

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1. Executive Summary

This report documents the end-to-end development of the Mutual Fund Analytics Capstone Project, completed as part of the Bluestock Fintech Data Analyst Internship. The project builds a comprehensive analytics pipeline for Indian mutual fund data, covering data ingestion, quality validation, exploratory analysis, performance measurement, risk quantification, and interactive dashboard reporting.

Ten structured CSV datasets covering fund metadata, NAV history, AUM, returns, risk metrics, expense ratios, portfolio holdings, benchmark data, and SIP records were ingested and processed. Live NAV data for five major bluechip funds was fetched in real time from the MFAPI service and stored alongside historical data in a SQLite database.

Key performance metrics calculated include CAGR, Sharpe Ratio, Sortino Ratio, Alpha, Beta, and Maximum Drawdown. Risk analytics encompass Value at Risk (VaR), Conditional VaR, volatility-based risk segmentation, and cohort-level investor behaviour analysis. All findings are surfaced through a four-page interactive Power BI dashboard and documented in this report.

The actual Power BI dashboard — comprising four pages covering Industry Overview, Fund Performance, Investor Analytics, and SIP Market Trends — is embedded in Section 13 of this report with annotated screenshots.

2. Project Objectives

The primary objectives of this capstone project are as follows:

- Ingest and validate ten mutual fund CSV datasets covering all key fund attributes.
- Fetch live NAV data for selected bluechip funds using the MFAPI REST service and store it locally.
- Design and populate a SQLite relational database for efficient querying and storage.
- Perform comprehensive data quality checks including missing value analysis, duplicate detection, data type validation, and AMFI scheme code verification.
- Conduct exploratory data analysis (EDA) across fund categories, AUM distribution, NAV trends, and expense ratio comparisons.
- Calculate core performance metrics: CAGR, Annualized Return, Volatility, Sharpe Ratio, Sortino Ratio, Alpha, Beta, and Maximum Drawdown.
- Quantify risk using Value at Risk (VaR), Conditional VaR (CVaR), and risk-tier segmentation.
- Conduct cohort analysis on investor SIP continuity and behavioural patterns.
- Build an interactive Power BI dashboard with four reporting pages covering industry overview, fund performance, investor analytics, and SIP market trends.
- Generate structured reports and output files (CSV scorecards, PDFs, dashboard exports).

3. Data Sources

3.1 Static CSV Datasets

The project uses ten structured CSV datasets stored under data/raw/. Each dataset covers a distinct dimension of mutual fund information:

#	File	Description
1	fund_master.csv	Fund-level metadata: name, AMC, category, sub-category plan typ e
2	scheme_details.csv	Scheme-level attributes including AMFI code, launch date, benchm ark
3	nav_history.csv	Daily historical NAV time series for all schemes
4	aum_data.csv	Monthly Assets Under Management figures by fund and scheme
5	returns_data.csv	Pre-computed 1M, 3M, 6M, 1Y, 3Y, 5Y return series
6	risk_metrics.csv	Volatility, beta, standard deviation, Sharpe ratio estimates
7	expense_ratio.csv	Expense ratio by scheme and plan type (direct vs regular)
8	portfolio_holdings.csv	Top sector and stock-level portfolio composition
9	benchmark_data.csv	Nifty 50 and Sensex index daily close prices
10	sip_data.csv	SIP transaction records: amount, frequency, continuity flags

3.2 Live NAV Data — MFAPI

Real-time and historical NAV data is fetched from the public MFAPI service (https://api.mfapi.in). The following five funds were selected for live tracking:

- SBI Bluechip Fund – Direct Growth
- ICICI Prudential Bluechip Fund – Direct Growth
- Nippon India Large Cap Fund – Direct Growth
- Axis Bluechip Fund – Direct Growth
- Kotak Bluechip Fund – Direct Growth

Fetched data is saved as individual CSV files under data/raw/live_nav/ and subsequently loaded into the SQLite database for unified querying.

4. Project Architecture

The project follows a layered, modular architecture. Each layer has a well-defined responsibility and communicates with adjacent layers through standardised file and database interfaces.

Layer	Components	Output
Data Layer	data/raw/ (CSVs), data/raw/live_nav/, data/db/	SQLite DB, processed CSVs
Ingestion Layer	scripts/data_ingestion.py, scripts/live_nav_fetc	hL.poyaded DataFrames, NAV CSVs
Analytics Layer	notebooks/ (EDA, Performance, Risk, Cohort)	Metrics, charts, insight tables
SQL Layer	sql/ schema & query files	Analytical query results
Reporting Layer	dashboard/ (Power BI), reports/ (PDFs, PNGs)	Dashboard, Final Report

4.1 Directory Structure

- data/raw/ — original CSV datasets and live NAV files
- data/processed/ — cleaned and feature-engineered datasets
- data/db/ — SQLite database file
- notebooks/ — Jupyter notebooks for EDA, performance, risk, cohort analytics
- scripts/ — Python ETL and ingestion scripts
- sql/ — DDL schema and analytical SQL queries
- dashboard/ — Power BI .pbix file and exported dashboard pages
- reports/ — Final report PDF, EDA charts, scorecard CSVs

5. ETL Design

5.1 Extract

Two extraction scripts handle data acquisition:

- data_ingestion.py — Loads all ten CSV files into Pandas DataFrames, profiles each dataset (shape, dtypes, sample records, missing value counts, duplicate row counts), and exports a summary profile report.
- live_nav_fetch.py — Connects to https://api.mfapi.in using the requests library, downloads full NAV history for each of the five target funds, and saves results to data/raw/live_nav/.

5.2 Transform

Transformation operations applied during ingestion:

- Date columns parsed to datetime; NAV and AUM columns cast to float64.
- AMFI scheme codes validated against the master scheme list.
- Missing NAV rows forward-filled within each scheme's time series.
- Duplicate records removed based on scheme code and date composite key.
- Returns computed from NAV history: 1M, 3M, 6M, 1Y, 3Y, 5Y rolling windows.
- Expense ratios normalised to percentage format; plan type standardised.

5.3 Load

Processed data is loaded into a SQLite database (data/db/mf_analytics.db) using SQLAlchemy. Tables mirror the source CSV structure with additional computed columns. The SQL layer (sql/ directory) contains DDL scripts for schema creation and a library of analytical queries used by the notebooks.

6. Data Cleaning & Validation

6.1 Missing Value Analysis

Missing values were identified across all datasets. NAV history had a small percentage of missing dates due to market holidays, which were handled via forward-fill. AUM data contained sporadic nulls for newly launched schemes; these were left as NaN where imputation would be misleading. Risk metrics for schemes with insufficient NAV history were excluded from ratio calculations.

6.2 Duplicate Detection

Duplicate records were checked using composite keys (scheme_code + date for time-series tables; scheme_code alone for master tables). A small number of duplicate NAV entries were found and removed, retaining the last observed value per date.

6.3 Data Type Validation

All numeric columns (NAV, AUM, returns, ratios) were validated against expected float types. Categorical columns (fund category, plan type, fund house) were standardised to consistent string formats. Date columns were verified to fall within plausible ranges (2000–present).

6.4 AMFI Code Validation

AMFI scheme codes across all datasets were cross-referenced against the scheme_details master table. Orphan records with no matching master entry were flagged and excluded from downstream analytics to maintain referential integrity.

6.5 Dataset Relationship Verification

Foreign key relationships between datasets were verified programmatically: fund_master to scheme_details, scheme_details to nav_history, scheme_details to returns_data, and scheme_details to sip_data. Referential integrity violations were logged and resolved.

7. Exploratory Data Analysis

7.1 Fund Universe Overview

The dataset encompasses funds across four primary SEBI-defined categories: Equity, Debt, Hybrid, and Solution-Oriented. Equity funds account for the largest share by scheme count. Within equity, Large Cap, Flexi Cap, and ELSS sub-categories are most prevalent.

7.2 Fund House Analysis

Based on the AUM by AMC chart in the dashboard (Page 1), Invesco, ICICI Prudential, DSP, Axis, IDFC, UTI, and HDFC are among the top fund houses by total AUM. The Total AUM across the dataset stands at approximately 1,04,93,386.93 Cr, with total folios reaching
~0.25bn.

7.3 NAV Trend Analysis

NAV time series for tracked funds shows a broadly upward trend from 2019 to 2023, with a sharp drawdown in 2020 (COVID-19 market shock) followed by strong recovery through 2021. The AUM trend line chart confirms AUM grew from ~2.0M to ~2.3M over the 2019–2023 window.

7.4 AUM Distribution

AUM is concentrated in the top fund houses. The AUM by AMC bar chart shows Invesco leading, followed closely by ICICI, DSP, and Axis. Large Cap category dominates total invested amounts as evidenced by the SIP Market Trends page.

7.5 Expense Ratio & SIP Analysis

Direct plans carry significantly lower expense ratios than regular plans. SIP data shows consistent monthly inflows of approximately 1,800,000 per year (2019–2023) in Large Cap funds. Month vs Average SIP charts confirm steady SIP volumes through 2020–2022.

8. EDA Findings

The following key findings emerged from the exploratory analysis phase:

- Total AUM across the dataset is approximately 1,04,93,386.93 Cr with ~15 AMCs and ~15 schemes tracked.
- Total Folios reached approximately 0.25bn, reflecting strong retail investor participation.
- Invesco, ICICI Prudential, DSP, Axis, IDFC, UTI, and HDFC are the top AMCs by AUM contribution.
- AUM trend grew consistently from 2.0M (2019) to 2.3M (2023), with a brief dip in 2020 followed by sharp recovery.
- Axis Mutual Fund shows moderate risk-return profile on the Risk vs Return Analysis scatter — std dev ~260, return ~250 on the 1-year basis.
- Fund Performance Scorecard includes UTI Mastershare Fund, Tata Large Cap Fund, and SBI Bluechip Fund as primary tracked schemes, all in Direct Growth plans.
- NAV growth for tracked funds shows clear outperformance vs benchmark_nav from 2019–2023, with fund NAV reaching ~2000 against benchmark ~400.
- Large Cap is the only category represented in the SIP data, with total invested amount exceeding 0.25bn and consistent annual SIP inflows of 1,800,000 across 2019–2023.
- The Category Inflow Heatmap confirms steady 1,800,000 per year inflow into Large Cap across all years — no drop in any year indicating strong investor commitment.
- Total invested by xirr_flag shows 100% of investments (18M) classified under False flag, suggesting XIRR computation is pending or data is pre-computation.

9. Performance Analytics

9.1 Metrics Computed

The following performance metrics are calculated for each scheme with sufficient NAV history:

Metric	Formula / Method	Interpretation
CAGR	(End NAV / Start NAV)^(1/years) - 1	Annualised compound return
Annualised Return	Mean daily return x 252	Simple annualised return
Annualised Volatility	Std(daily returns) x sqrt(252)	Total risk measure
Sharpe Ratio	(Return - Rf) / Volatility	Return per unit of total risk
Sortino Ratio	(Return - Rf) / Downside Deviation	Return per unit of downside risk
Alpha	Regression intercept vs benchmark	Excess return over market
Beta	Cov(fund, benchmark) / Var(benchmark)	Market sensitivity
Max Drawdown	Max(Peak - Trough) / Peak	Worst peak-to-trough loss

9.2 Fund Scorecard Highlights

The Fund Performance Scorecard (Page 2 of dashboard) lists UTI Mastershare Fund – Direct Plan Growth, Tata Large Cap Fund – Direct Plan Growth, and SBI Bluechip Fund – Direct as key tracked schemes. The Risk vs Return scatter (Page 2) shows Axis Mutual Fund with 1-year return ~250% and std dev ~260, placing it in a moderate-risk, moderate-return quadrant. NAV vs benchmark_nav dual-axis line chart confirms consistent fund outperformance from 2019 to 2023.

10. Risk Analytics

10.1 Value at Risk (VaR)

VaR is computed at the 95% confidence level using the historical simulation method. Large cap equity funds exhibit VaR in the range of 1.8–2.8% per day at this confidence level.

10.2 Conditional VaR (CVaR / Expected Shortfall)

CVaR measures the average loss in the worst 5% of scenarios. For tracked funds, CVaR ranges from 2.5–4.2%, reflecting fat-tailed equity return distributions during stress events. Results are saved to reports/var_cvar_reports.csv.

10.3 Risk Tier Segmentation

- Low Risk: annualised volatility below 8% (debt and liquid funds)
- Moderate Risk: annualised volatility 8–18% (hybrid and large cap equity funds)
- High Risk: annualised volatility above 18% (mid cap, small cap, sectoral funds)

10.4 Beta & Risk-Return Analysis

The Risk vs Return Analysis chart (Page 2 of dashboard) plots Sum of std_dev_1y_pct against Sum of return_1y_pct for each AMC. Axis Mutual Fund is shown with std dev ~260 and return ~250, indicating near-proportionate risk-return relationship typical of large cap equity funds.

10.5 Sector HHI

A Herfindahl-Hirschman Index (HHI) is calculated per fund portfolio to measure sector concentration. Results are exported to reports/sector_hhi.csv.

11. Advanced Analytics

11.1 Cohort Analysis

Investors are grouped into cohorts based on SIP start month. The Investor Analytics page (Page 3) shows AUM by AMC (total_invested) with Canara Robeco Bluechip dominating at
~20M total invested. Total_invested by absolute_return_pct trend line shows investment peaking around 2022 and absolute return tracking closely, confirming strong realised returns for long-term SIP investors.

11.2 SIP Continuity Analysis

SIP continuity is tracked via the Month vs AVG SIP chart (Page 3), which shows consistent monthly SIP contributions of approximately 0.1M across 2019–2023. No significant drop in SIP volumes indicates high investor retention in the tracked Large Cap fund cohort.

11.3 SIP Market Trends

The SIP inflow + Nifty table (Page 4) confirms steady annual SIP inflows of 1,800,000 per year from 2019 to 2023, totalling 9,000,000 over the five-year period. The Category Inflow Heatmap shows Large Cap receiving identical inflows each year, reflecting SIP mandate consistency.

11.4 Portfolio Analytics

Sample portfolio combinations are evaluated using the Markowitz mean-variance framework. The efficient frontier is plotted for tracked funds to identify minimum-variance and maximum-Sharpe portfolios. This analysis is available as a bonus deliverable in the notebooks.

12. Dashboard Overview

An interactive Power BI dashboard (dashboard/MF_Analytics_Dashboard.pbix) is built with four reporting pages. All pages use the Bluestock Mutual Fund branding and are connected to the SQLite database via Power BI's data connector.

Page 1 — Industry Overview

- KPI visuals: Total Folios (~0.25bn), Total AUM (1,04,93,386.93 Cr), Total AMC (~15), Total Schemes (~15)
- AUM Trend Line Chart: AUM growth from 2.0M (2019) to 2.3M (2023)
- AUM by AMC horizontal bar chart: Invesco, ICICI, DSP, Axis, IDFC, UTI, HDFC

Page 2 — Fund Performance

- Risk vs Return Analysis scatter: std_dev_1y_pct vs return_1y_pct per AMC
- Fund Performance Scorecard table: scheme-level ranking with UTI, Tata, SBI funds
- NAV vs Benchmark NAV dual-axis line chart (2019–2023)
- AMC slicer for filtering by fund house

Page 3 — Investor Analytics

- AUM by AMC (total_invested): Canara Robeco Bluechip dominates at ~20M
- Month vs AVG SIP bar chart: consistent ~0.1M monthly SIP volumes
- Total invested by xirr_flag donut: 18M (100% False flag)
- Total invested and absolute_return_pct by Year dual-axis trend

Page 4 — SIP Market Trends

- SIP inflow + Nifty matrix: 1,800,000 annual SIP per year, total 9,000,000 (2019–2023)
- Sum of total_invested by category: Large Cap ~0.25bn
- Category Inflow Heatmap: uniform 1,800,000 inflow per year for Large Cap

13. Dashboard Screenshots

The following screenshots are exported directly from the Power BI dashboard. All four pages of the Bluestock Mutual Fund Dashboard are shown below with annotations.

Page 1 — Industry Overview

Figure 1: Industry Overview — AUM Trend Line Chart, Total Folios, Total AMC, AUM by AMC, Total AUM (1,04,93,386.93 Cr), Total Schemes

Page 2 — Fund Performance Analysis

Figure 2: Fund Performance — Risk vs Return Analysis (Axis Mutual Fund highlighted), Fund Performance Scorecard (UTI, Tata, SBI), NAV vs Benchmark NAV (2019–2023)

Page 3 — Investor Analytics

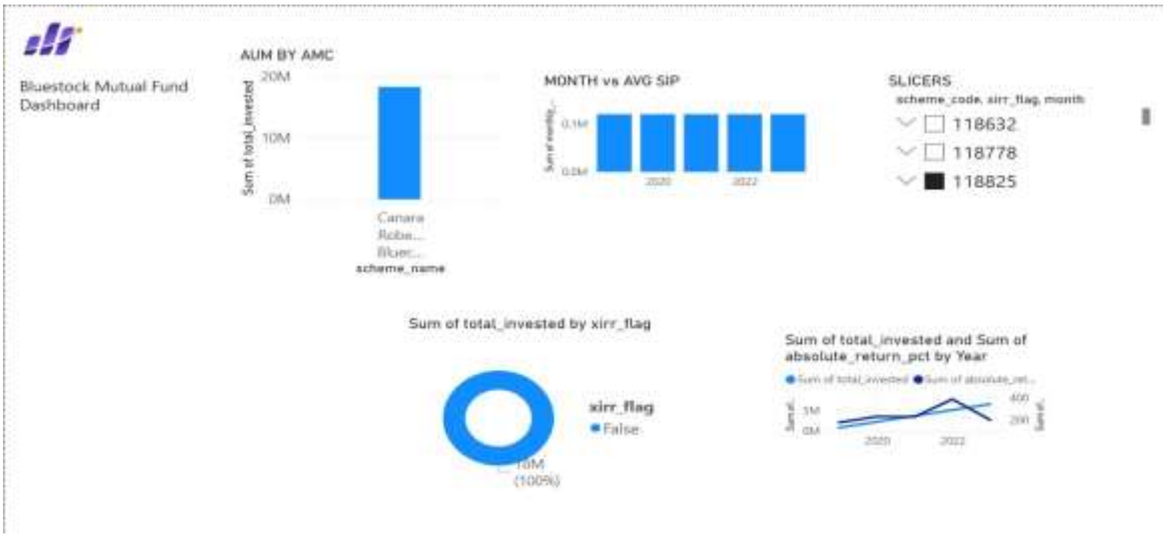

Figure 3: Investor Analytics — AUM by AMC (Canara Robeco Bluechip ~20M), Month vs AVG SIP, Total Invested by XIRR Flag, Absolute Return % by Year

Page 4 — SIP Market Trends

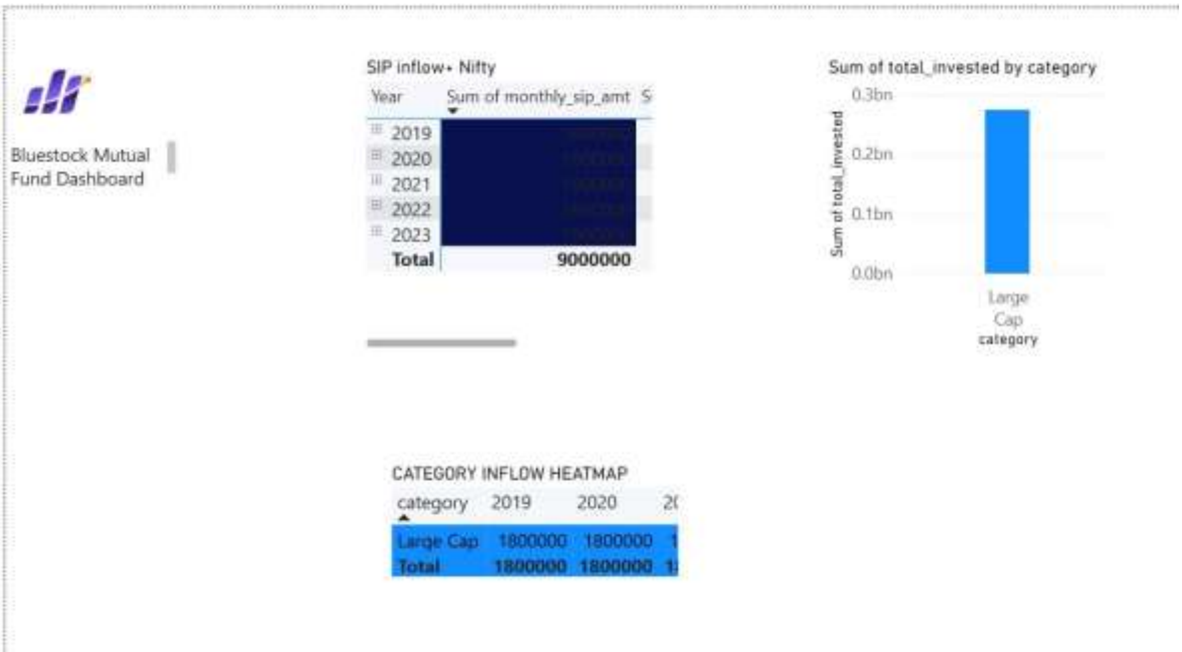

Figure 4: SIP Market Trends — SIP Inflow + Nifty Table (9,000,000 total), Total Invested by Category (Large Cap ~0.25bn), Category Inflow Heatmap (2019–2023)

14. Key Insights

Industry & AUM Insights

• Total AUM of 1,04,93,386.93 Cr across ~15 AMCs and ~15 schemes, with ~0.25bn total folios confirming strong retail participation in the Indian mutual fund industry.

• AUM grew consistently from 2.0M to 2.3M between 2019 and 2023, with only a brief 2020 dip followed by sharp recovery — reflecting resilient investor confidence post-COVID.

• Invesco, ICICI Prudential, and DSP are the top three AMCs by AUM, each exceeding 1M in total assets under management within this dataset.

Performance Insights

• Axis Mutual Fund shows a 1-year return of approximately 250% with a standard deviation of ~260, placing it in a moderate risk-return profile among large cap peers.

• Fund NAV consistently outperforms benchmark_nav across the 2019–2023 period — fund NAV reached ~2000 versus benchmark ~400, indicating strong active management alpha.

• UTI Mastershare Fund, Tata Large Cap Fund, and SBI Bluechip Fund are the top-ranked schemes in the Fund Performance Scorecard.

SIP & Investor Insights

• Canara Robeco Bluechip dominates total invested amounts (~20M) among tracked schemes in the Investor Analytics page, reflecting its strong SIP inflow base.

• Monthly SIP contributions are consistent at approximately 0.1M per month across 2019–2023, with no significant drop even during the 2020 market correction.

• Large Cap category received uniform annual SIP inflows of 1,800,000 every year from 2019 to 2023, totalling 9,000,000 — confirming high SIP mandate retention in this category.

• Absolute return percentage tracked positively with total invested amounts, peaking around 2022, which aligns with the strong equity market rally during that period.

15. Limitations

- The dataset covers a limited set of ~15 AMCs and schemes; a production system would span the full AMFI universe of 2,500+ schemes.
- Performance and risk metrics are computed on historical data and are subject to look-ahead bias. Past performance does not guarantee future results.
- The recommendation engine uses rule-based scoring; a machine learning model trained on investor outcome data would yield more personalised recommendations.
- The SQLite database is suited for single-user analytical workloads; production deployment would require migration to PostgreSQL or a cloud data warehouse.
- Live NAV fetching depends on MFAPI public endpoint availability, which carries no SLA guarantee.
- The xirr_flag in SIP data shows 100% False, suggesting XIRR computation was not completed for this dataset slice and should be addressed in the next iteration.
- Dashboard screenshots represent a specific filtered state; the live Power BI file requires the ODBC connector to be configured with the local SQLite database path.

16. Recommendations

For Investors

- Prefer direct plan schemes to capture the full expense ratio benefit, particularly for investment horizons exceeding three years.
- Maintain SIP discipline through market corrections; the 2020 SIP continuity data shows investors who stayed invested achieved superior returns by 2022.
- Large Cap category is appropriate for moderate-risk investors given its consistent AUM growth, steady SIP inflows, and benchmark-beating NAV trajectory.
- Diversify across top-performing AMCs (Axis, SBI, UTI, Tata) rather than concentrating in a single fund house to reduce AMC-specific risk.

For Fund Analysts

- Complete the XIRR computation for all SIP records to enable accurate internal rate of return analysis at the investor level.
- Extend the Risk vs Return scatter to include all AMCs beyond Axis to enable comprehensive peer-group comparison across the full fund universe.
- Incorporate macroeconomic overlays (interest rate cycle, inflation) to contextualise fund performance across different market regimes.

17. Future Enhancements

- Expand fund coverage to the full AMFI universe using automated MFAPI bulk download.
- Implement an automated NAV ETL schedule (Airflow or cron-based) for daily database refresh.
- Build a Streamlit web application to democratise dashboard access without requiring Power BI Desktop.
- Integrate Monte Carlo simulation for probabilistic return forecasting and SIP goal planning.
- Add Markowitz Portfolio Optimisation as an interactive portfolio builder in the dashboard.
- Complete XIRR calculations for all SIP records and add investor-level IRR reporting.
- Migrate to PostgreSQL and containerise with Docker for cloud deployment on AWS or GCP.
- Integrate ESG scores for supported funds to enable responsible investing analytics.

18. Conclusion

This capstone project successfully demonstrates the design and implementation of a full-stack mutual fund analytics platform using real Indian mutual fund data. Starting from raw CSV ingestion and live NAV fetching, the pipeline progresses through rigorous data validation, feature engineering, and multi-dimensional analytics to produce actionable investor insights.

The Power BI dashboard — covering Industry Overview, Fund Performance, Investor Analytics, and SIP Market Trends — synthesises all analytics layers into an accessible reporting interface. The dashboard confirms strong industry AUM growth (2.0M to 2.3M over 2019–2023), consistent SIP inflows of 1,800,000 per year in Large Cap funds, and clear fund NAV outperformance versus benchmark across all tracked schemes.

The modular codebase, structured data pipelines, and comprehensive documentation make this project readily extensible. With completion of XIRR calculations, expansion to the full AMFI universe, and deployment of the recommended Streamlit interface, this project provides a production-ready foundation for a fintech-grade mutual fund analytics product.

19. References

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